

Calm Months, Crash Months: Why Volatility Management Pays for Momentum and Not the Market

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Abstract

Built with only past data, capped leverage and trading costs, volatility management improves momentum, from a Sharpe ratio of 0.43 to 0.94 over July 1973 to August 2026, but none of the other Fama-French factors. We explain why. Sorting stocks on past returns loads momentum on the market in proportion to the market's own return over the formation window. Using Tweedie's formula, we derive this beta in closed form from two cross-sectional quantities, the dispersion of stock betas and the density of past returns at the portfolio breakpoints, and test it on CRSP with no fitted parameter. Against realized betas from 1932 to 2025, the prediction has a slope of 0.92 (standard error 0.19) and 79% of the explanatory power of an in-sample regression: it captures how momentum's beta moves, while the realized level sits 0.20 below the prediction. Months after market declines are 29% of the sample and supply 71% of the covariance through which timing adds to momentum's return. If the market's premium rises with its variance, the same logic implies that the market should not be volatility-timed; in our sample it is not improved by it.

Keywords: volatility-managed portfolios, momentum, time-varying beta, Tweedie's formula, factor timing

JEL: G11, G12, G14

1 Introduction

Moreira and Muir (2017) show that scaling a factor's exposure by the inverse of its recent variance raises the Sharpe ratio of many equity factors. Their evidence is a spanning regression that uses full-sample moments no investor had. In real time the gains mostly disappear (Cederburg et al., 2020; Liu et al., 2019; Barroso and Detzel, 2021), and momentum is the consistent survivor, in line with the crash evidence of Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016). Why momentum, and why not the market, for which the rule was first proposed?

We answer in three steps. First, we rebuild the six most widely used equity factors so that every position uses only past data, leverage is capped and costs are paid; only momentum improves. An

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exact decomposition shows why: timing adds return only when the factor earns less in turbulent months, and among the six factors that holds for momentum alone. Second, we derive where momentum’s turbulent-month losses come from. Momentum is long past winners and short past losers, so after a market decline its short leg holds the high-beta stocks and the factor is short the market (Grundy and Martin, 2001; Daniel and Moskowitz, 2016). Proposition 1 gives this beta in closed form from two quantities measured in the cross-section of stocks, and we test it on CRSP without fitting any parameter. Third, we show what volatility timing should time: only the part of a premium that is not paid for market risk. If the market’s premium rises with its variance (Merton, 1980), the market should not be timed at all.

The contribution is the formula and its parameter-free test. The sign of momentum’s conditional beta is known (Grundy and Martin, 2001; Daniel and Moskowitz, 2016); we derive its size and test it without estimating anything from momentum’s returns. The main test was specified in a plan committed to the code repository before the CRSP data were downloaded, and every later test before it was run (`paper/PLAN_mechanism.md`).

2 Data and real-time method

We use the market, size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum factors of Fama and French (2015) and Carhart (1997) from the Kenneth R. French Data Library, July 1963 to August 2026. Monthly returns f_m come from the monthly files; daily returns r_d serve only to measure risk. The variance signal uses every daily return with a one-month half-life,

$$\hat{\sigma}_d^2 = \lambda \hat{\sigma}_{d-1}^2 + (1 - \lambda) r_d^2, \quad \lambda = 2^{-1/21}, \quad S_m = 21 \hat{\sigma}_{d(m)}^2, \quad (1)$$

with $d(m)$ the last trading day of month m . This learning rate was chosen for momentum on 1973 to 2016 data among 24 pre-registered variants (`paper/PLAN_adaptive.md`). The canonical prior-month realized variance of Moreira and Muir (2017) is reported alongside.

The exposure held in month m is set at the end of month $m - 1$:

$$w_m = \min\left(\frac{c_{m-1}}{S_{m-1}}, 1.5\right), \quad c_{m-1} = \frac{\text{sd}(f_1, \dots, f_{m-1})}{\text{sd}(f_s/S_{s-1} : s \leq m-1)}, \quad (2)$$

an expanding-window version of Moreira and Muir’s constant with their higher cap. The first 120 months are a burn-in, so positions start in July 1973 (638 months). Exposure changes cost 14 basis points per unit, their highest cost level: $f_m^\sigma = w_m f_m - \kappa |w_m - w_{m-1}|$. Sharpe ratios are compared with the studentized circular block bootstrap of Ledoit and Wolf (2008) (12-month blocks, 4,999 resamples), Holm-adjusted across the six factors (Holm, 1979).

Why timing can help or hurt. The mean managed return decomposes exactly as $\overline{f^\sigma} = \bar{w} \bar{f} + \widehat{\text{cov}}(w_m, f_m) - \kappa |\overline{\Delta w}|$. A constant exposure $\bar{w}$ has the original factor’s Sharpe ratio, so whenever $\bar{f} > 0$,

$$\frac{\text{SR}(f^\sigma)}{\text{SR}(f)} = \underbrace{\left(1 + \frac{\widehat{\text{cov}}(w_m, f_m) - \kappa |\overline{\Delta w}|}{\bar{w} \bar{f}}\right)}_{M: \text{ return channel}} \times \underbrace{\frac{\bar{w} \text{sd}(f)}{\text{sd}(f^\sigma)}}_{R: \text{ risk channel}}. \quad (3)$$

R is the risk that volatility timing removes. M is the return it adds or gives up: it exceeds one only if the factor earns more in calm months, when the rule holds more of it.

3 Only momentum improves

Factor	Managed factor, net				Real-time combination, net			Largest loss, equal volatility	
	Original	Canonical	Daily-data	p (Holm)	Original	Combination	p (Holm)	Original	Managed
Market	0.51	0.51	0.48	1.000	0.57	0.43	0.506	-68	-42
Size (SMB)	0.21	0.12	0.13	1.000	0.10	0.02	0.137	-77	-85
Value (HML)	0.32	0.23	0.27	1.000	0.24	0.18	0.343	-79	-67
Profitability (RMW)	0.38	0.47	0.45	1.000	0.47	0.35	0.545	-51	-46
Investment (CMA)	0.45	0.34	0.41	1.000	0.37	0.30	0.506	-31	-47
Momentum	0.43	0.90	0.94	0.023	0.42	0.80	0.343	-76	-23

Table 1: Annualized Sharpe ratios, July 1973 to August 2026, exposure capped at 1.5, net of 14 basis points per unit of exposure change. Canonical is the prior-month realized-variance rule; daily-data is equation (1). The combination is the real-time mix of original and managed factor of Cederburg et al. (2020), from July 1978. p -values test equal Sharpe ratios against the original factor, Holm-adjusted across the six factors; bold marks adjusted $p < 0.05$. Largest loss is the maximum peak-to-trough decline of summed monthly returns, in percentage points, at the original factor’s volatility.

Managed momentum raises the Sharpe ratio from 0.43 to 0.94 after costs ($p = 0.004$, Holm-adjusted 0.023; Table 1), and its largest loss at equal volatility falls from 76 to 23 percentage points. No other factor improves; the market’s Sharpe ratio falls from 0.51 to 0.48. No real-time combination beats its original factor after costs. After the published samples end the gain is smaller, 0.21 to 0.44 over 116 months against 0.48 to 1.02 before, and the change between the periods is not significant ($p = 0.242$). Across variance signals, block lengths, caps, scaling rules and start dates, managed momentum’s full-sample Sharpe ratio ranges from 0.78 to 0.97 and always differs from the original ($p \leq 0.014$).

Factor	Period	$\bar{w}$	$\bar{f}$	$\bar{w}\bar{f}$	$\widehat{\text{cov}}(w, f)$	Cost	$\bar{f}^\sigma$	M	R
Market	1973 to 2016	0.74	6.9	5.1	-0.2	0.2	4.6	0.91	1.13
	2017 to 2026	0.73	13.1	9.6	-2.8	0.3	6.5	0.67	1.14
Size (SMB)	1973 to 2016	0.59	3.2	1.9	-0.3	0.2	1.4	0.71	1.10
	2017 to 2026	0.43	-2.6	-1.1	-0.9	0.1	-2.2	n/a	0.90
Value (HML)	1973 to 2016	0.79	4.5	3.6	-0.9	0.2	2.4	0.67	1.18
	2017 to 2026	0.27	-1.7	-0.4	-1.0	0.1	-1.5	n/a	1.08
Profitability (RMW)	1973 to 2016	0.84	3.4	2.8	-0.4	0.2	2.3	0.79	1.52
	2017 to 2026	0.40	2.4	1.0	-0.4	0.1	0.4	0.45	1.33
Investment (CMA)	1973 to 2016	0.95	4.2	4.0	-0.8	0.2	2.9	0.73	1.22
	2017 to 2026	0.50	-1.0	-0.5	-1.1	0.2	-1.8	n/a	1.11
Momentum	1973 to 2016	0.82	7.3	6.0	2.5	0.2	8.2	1.37	1.56
	2017 to 2026	0.35	2.9	1.0	1.0	0.1	1.9	1.87	1.13

Table 2: Exact decomposition of the daily-data rule, equation (3). $\bar{f}$, $\bar{w}\bar{f}$, the covariance, the cost and $\bar{f}^\sigma$ are annualized, in percent. M and R are the return and risk channels; their product is the ratio of managed to original Sharpe ratios. M is not reported when $\bar{f} \leq 0$.

Table 2 shows the reason. Momentum’s covariance between exposure and return is positive

in both periods, so its return channel exceeds one ($M = 1.37$ before 2017). Every other factor's covariance is negative: the rule scales them down in the months they pay. Figure 1 shows the split directly: in both periods momentum earns more in calm months than in turbulent ones, and every other factor the reverse.

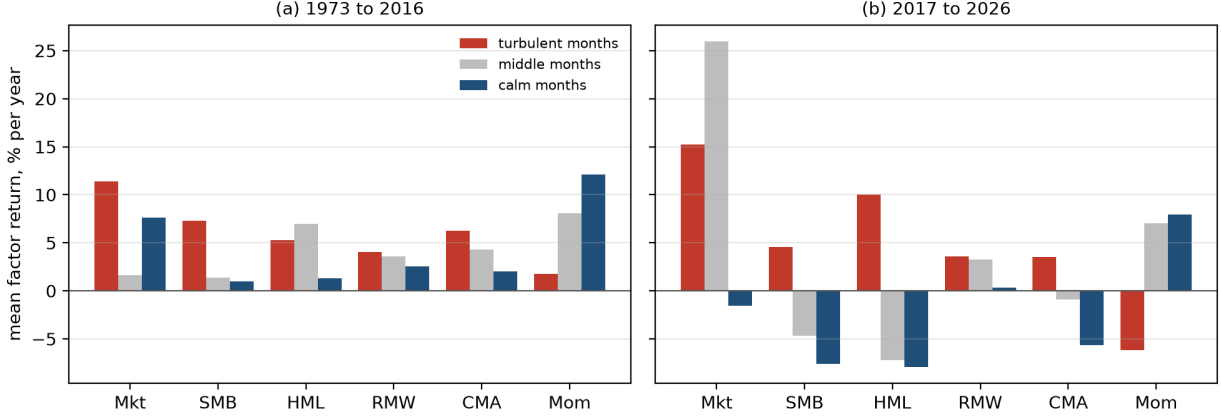


Figure 1: Annualized mean factor return by tercile of the real-time managed exposure (daily-data rule), before and after the published samples end. Turbulent months are the third with the lowest exposure, that is, the highest prior variance.

For the market, M is 0.91 before 2017 and 0.67 after. The next two sections explain both signs.

4 Momentum's formation beta

Proposition 1. Let stock i 's market beta be $b_i = \bar{b} + \sigma_b \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, 1)$, and its formation-window log return $r_i = b_i R_F + u_i$, where u_i is independent of b_i and has any distribution, and R_F is the coefficient on beta, to first order $\sum_t R_{M,t}(1 - R_{M,t} - r_{f,t})$ over the window for market excess returns $R_{M,t}$. Winners are the stocks with $r_i \geq c_{hi}$ (a share q_{hi}), losers those with $r_i \leq c_{lo}$ (a share q_{lo}). If betas persist into the holding month, the equal-weighted winners-minus-losers beta is

$$\beta_{WML} = \sigma_b^2 R_F \left[\frac{\phi(c_{hi})}{q_{hi}} + \frac{\phi(c_{lo})}{q_{lo}} \right], \quad (4)$$

where ϕ is the cross-sectional density of formation returns.

Proof. Write $r_i = \theta_i + e_i$ with $e_i = (b_i - \bar{b})R_F \sim \mathcal{N}(0, \sigma_b^2 R_F^2)$ independent of $\theta_i = \bar{b}R_F + u_i$. Tweedie's formula (Efron, 2011) gives $\mathbb{E}[e_i | r_i] = -\sigma_b^2 R_F^2 (\log \phi)'(r_i)$, so $\mathbb{E}[b_i | r_i] = \bar{b} - \sigma_b^2 R_F (\log \phi)'(r_i)$. Averaging over winners, $\mathbb{E}[-(\log \phi)'(r) | r \geq c_{hi}] = q_{hi}^{-1} \int_{c_{hi}}^{\infty} -\phi'(r) dr = \phi(c_{hi})/q_{hi}$; over losers the same integral gives $-\phi(c_{lo})/q_{lo}$. Subtract. $\square$

The sign comes from the market's own formation return: after a bull market momentum is long the market, after a bear market short it. With normal u_i and equal tail shares, (4) is linear in R_F near zero, the relation Grundy and Martin (2001) estimate by regression, and saturates after extreme markets. Its size depends on how many stocks sit near the breakpoints, so heavier-tailed

idiosyncratic returns, which put more stocks near the 30th and 70th percentiles at the same variance, raise it.

Test. We measure both inputs in CRSP (common stocks on NYSE, AMEX and NASDAQ, monthly, to December 2025). Our value-weighted rebuild of French’s momentum correlates 0.9989 with his over 1,181 months. For the portfolio formed at the end of month m , each stock’s beta $\hat{b}_i$ is estimated over months $m - 71$ to $m - 12$, before the formation window starts, so no beta uses the returns being sorted on. σ_b^2 is the cross-sectional variance of $\hat{b}_i$ minus the mean of their squared standard errors, and ϕ at the NYSE breakpoints comes from the quantile-spacing estimator with the bandwidth of Hall and Sheather (1988). Both are computed within each of French’s two size groups and averaged. The test regression is

$$\text{MOM}_{m+1} = a + c \hat{\beta}_m R_{M,m+1} + d R_{M,m+1} + \varepsilon_{m+1}, \quad (5)$$

with Newey-West standard errors (12 lags). The prediction is exact if $c = 1$ and $d = 0$. On simulated markets in which (4) holds, with normal and t_4 idiosyncratic returns, the procedure returns c of 1.04 on average and the joint test rejects at its nominal rate.

Sample	Portfolio	c (s.e.)	d (s.e.)	$p: c = 1, d = 0$	ΔR^2	Share of in-sample	Months
1932 to 2025	Equal-weighted	0.88 (0.20)	-0.22 (0.06)	0.000	0.21	0.75	1122
1932 to 2025	French’s momentum	0.92 (0.19)	-0.20 (0.06)	0.000	0.24	0.79	1122
1963 to 2025	Equal-weighted	1.00 (0.17)	-0.16 (0.06)	0.014	0.14	0.79	750
1963 to 2025	French’s momentum	1.08 (0.17)	-0.15 (0.06)	0.026	0.16	0.81	750

Table 3: Parameter-free test of Proposition 1. ΔR^2 is the increase in R^2 over a regression on $R_{M,m+1}$ alone; the share compares it with the increase from an in-sample regression that fits the slope on $R_F R_{M,m+1}$ (Grundy and Martin, 2001). Equal-weighted is the theory’s portfolio, French’s momentum the traded factor.

The slope matches the prediction (Table 3 and Figure 2). With no fitted parameter, c is 0.92 (standard error 0.19) for French’s momentum over 1932 to 2025 and 1.08 (0.17) from 1963; for the theory’s equal-weighted portfolio from 1963 it is 1.00. Adding the prediction raises the R^2 of a regression of momentum on the market by 24 percentage points, 79% of the rise from an in-sample regression that fits the slope. The level differs: realized betas sit 0.20 below the prediction over the full sample and 0.15 below from 1963 (d in Table 3), so a joint test of an exact prediction rejects. Figure 3 follows both over time: the prediction tracks the large swings, including the negative betas after the declines of 1937 and 2008 (correlation 0.51 across overlapping 12-month windows), while the realized series is noisier.

A plausible source of the gap is volatility drag. Compounding lowers the formation return of volatile stocks, so if high-beta stocks are also more volatile, the loser leg holds high-beta stocks even when $R_F = 0$, which would lower momentum’s beta at every level of the prediction; simulated markets built that way produce a gap of the same sign. Measuring it requires the joint distribution of betas and idiosyncratic volatility across stocks, which we leave to future work. Betas estimated before the formation window, averaged over winners and losers, move with the prediction (correlation 0.71, slope 0.55), as expected if stock betas drift between the estimation window and the formation year.

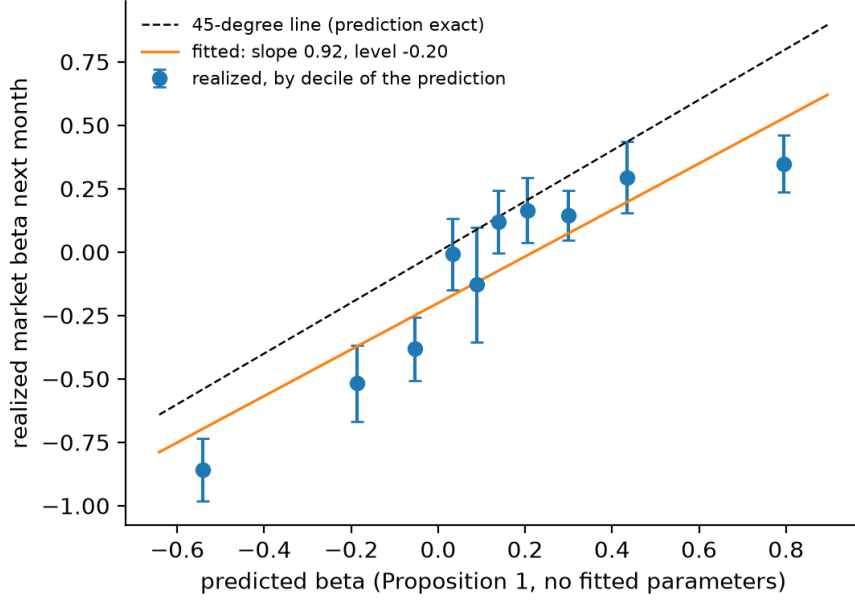


Figure 2: Realized market beta of French’s momentum in the month after formation, estimated within each decile of the predicted beta (4), holding months from July 1932 to December 2025, with 95% intervals. The dashed line is the exact prediction; the solid line is the fitted relation from Table 3.

5 What volatility timing should time

Proposition 2. *If the market premium is proportional to market variance, $\lambda_t = \kappa F_t^M$ (Merton, 1980), a factor with conditional expected return $\alpha + \beta_t \lambda_t$ and variance F_t has mean-variance weight*

$$w_t^* = \frac{1}{A} \left[\frac{\alpha}{F_t} + \kappa \beta_t \frac{F_t^M}{F_t} \right]. \quad (6)$$

Inverse-variance timing is optimal only for α , the part of the premium not paid for market risk. For the market ($\alpha = 0$, $\beta = 1$, $F = F^M$) the optimal weight is constant: an inverse-variance rule cuts exposure in exactly the months that the premium rises. The market’s return channel below one in Table 2 is consistent with that. For momentum, Proposition 1 makes β_t negative after market declines, so the market-related part of its expected return is most negative when market variance is high, in the months an inverse-variance rule already avoids.

The data agree. Over 629 holding months from July 1973 to November 2025, the months after market declines ($\hat{\beta} < 0$) are 29% of the sample and contribute 71% of the covariance between momentum’s managed exposure and its return. In those months momentum earns 1.6% a year against 8.6% otherwise, and the rule holds 0.56 times the factor against 0.82. Volatility management does not need to forecast this. A mean forecast built from $\hat{\beta}_m$ alone does not beat the historical mean (out-of-sample R^2 of -0.30% before 2017, Clark-West $p = 0.154$; Clark and West, 2007); the gain works through scaling exposure, not through a return forecast.

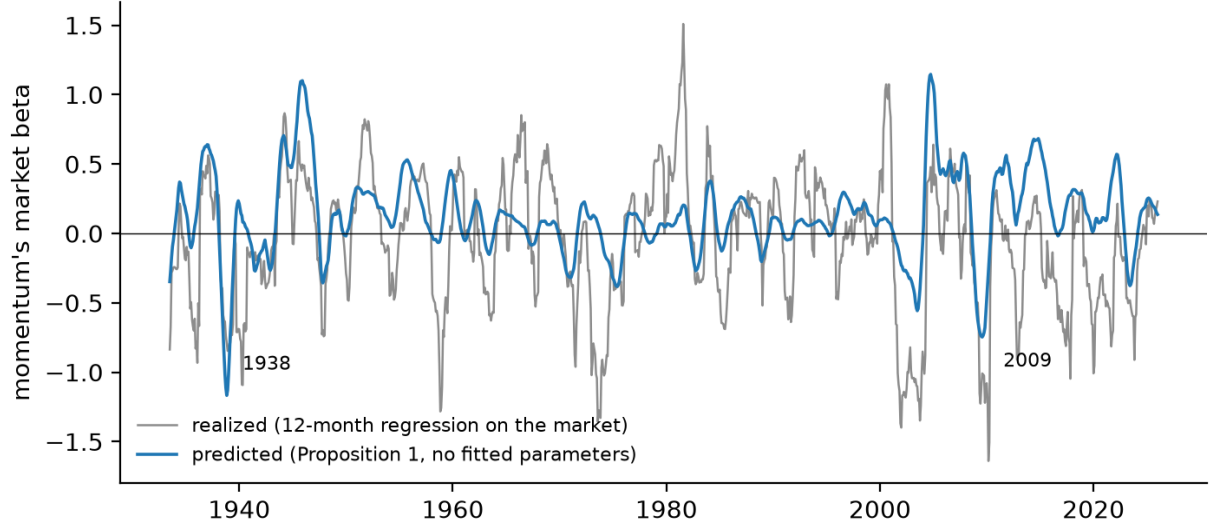


Figure 3: Momentum’s predicted beta (4), averaged over 12 consecutive holding months, and its realized beta, the slope of French’s momentum on the market over the same months, July 1932 to December 2025. A 12-month realized beta has a sampling error of about 0.22.

6 Conclusion

Volatility management pays for momentum and for none of the other five factors, in real time and after costs. The reason is how momentum is built. Sorting on past returns makes its market beta a function of the market’s own recent return, and Proposition 1 gives that function in closed form. With no fitted parameter it predicts how momentum’s beta moves, while its average level sits about 0.20 lower. After market declines the beta turns negative, and those months carry most of the covariance through which timing adds to momentum’s return. If the market’s premium rises with its variance, timing the market by variance gives up the premium; the market should not be volatility-timed, and in our sample it gains nothing.

Two extensions follow. Explaining the level of momentum’s beta needs a model of how betas and idiosyncratic volatility co-move across stocks. Net figures charge costs on changes in exposure, as in Moreira and Muir (2017), not on the trading inside each factor portfolio.

Data availability. Factor returns are available from the Kenneth R. French Data Library; stock returns are from CRSP through WRDS and cannot be redistributed. Code and the pre-registration plans are available from the author on request.

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